Se cargan los datos

```
import pandas as pd
import numpy as np
import statsmodels.api as sm
from sklearn.metrics import r2_score


# 1. Cargar

df = pd.read_csv("kc_house_data.csv")

print("Dimensiones:", df.shape)
df.head()
```

```
Dimensiones: (21613, 21)
```

	id	date	price	bedrooms	bathrooms	sqft_living	sqft_lot	floors	waterfront	view	...	grade	sqft_above	sqft_basement	yr_built	yr_ren
0	7129300520	20141013T000000	221900.0	3	1.00	1180	5650	1.0	0	0	...	7	1180	0	1955	
1	6414100192	20141209T000000	538000.0	3	2.25	2570	7242	2.0	0	0	...	7	2170	400	1951	
2	5631500400	20150225T000000	180000.0	2	1.00	770	10000	1.0	0	0	...	6	770	0	1933	
3	2487200875	20141209T000000	604000.0	4	3.00	1960	5000	1.0	0	0	...	7	1050	910	1965	
4	1954400510	20150218T000000	510000.0	3	2.00	1680	8080	1.0	0	0	...	8	1680	0	1987	

5 rows × 21 columns

Se establecen variables

```
# Variable dependiente a predecir
y = df['price']

# Variables independientes
X = df[[
    'bedrooms',
    'bathrooms',
    'sqft_living',
    'sqft_lot',
    'floors',
    'waterfront',
    'view',
    'condition',
    'grade',
    'sqft_basement',
    'yr_built',
    'yr_renovated',
    'lat',
    'long',
    'sqft_living15',
    'sqft_lot15'
]]
```

Se agrega intercepto, convierte a matrices y calcula los coeficientes

```
# 2. Agregar columna de unos

X_mat = np.column_stack((np.ones(len(X)), X))

# Convertir a matrices numpy
X_mat = np.array(X_mat)
y_mat = np.array(y)

# 3. Beta = (X'X)^(-1)X'Y


beta = np.linalg.inv(X_mat.T @ X_mat) @ X_mat.T @ y_mat

coeficientes = pd.DataFrame({
    'Variable':['Intercepto'] + list(X.columns),
    'Coeficiente':beta
})

coeficientes
```

Se encuentran las variables significativas

```
# Variables con p > 0.05

variables_significativas = modelo.pvalues[
    modelo.pvalues < 0.05
].index.tolist()

variables_significativas.remove('const')

print("Variables significativas:")
print(variables_significativas)
```

```
Variables significativas:
['bedrooms', 'bathrooms', 'sqft_living', 'sqft_lot', 'waterfront', 'view', 'condition', 'grade', 'sqft_basement', 'yr_built', 'yr_renovated', 'lat', 'lon
g', 'sqft_living15', 'sqft_lot15']
```

Bondad de ajuste

```
# 5. R²

r2 = r2_score(y, y_pred)

n = len(y)
k = X.shape[1]

r2_adj = 1 - ((1-r2)*(n-1)/(n-k-1))

print("R² =", round(r2,4))
print("R² Ajustado =", round(r2_adj,4))
```

```
R² = 0.6954
R² Ajustado = 0.6952
```

Ecuación final

```
ECUACIÓN FINAL:

Precio = -36922319.02 + (-34164.4966)*bedrooms + (42341.0922)*bathrooms + (179.6412)*sqft_living + (0.1272)*sqft_lot + (587854.2714)*waterfront + (49444.
0313)*view + (30998.1762)*condition + (97258.3187)*grade + (-33.2686)*sqft_basement + (-2453.0853)*yr_built + (21.5718)*yr_renovated + (561297.2535)*lat
+ (-117371.0868)*long + (27.3428)*sqft_living15 + (-0.3938)*sqft_lot15
```

Reporte

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                  price   R-squared:                       0.695
Model:                            OLS   Adj. R-squared:                  0.695
Method:                 Least Squares   F-statistic:                     3287.
Date:                Tue, 30 Jun 2026   Prob (F-statistic):               0.00
Time:                        08:29:59   Log-Likelihood:            -2.9476e+05
No. Observations:               21613   AIC:                         5.895e+05
Df Residuals:                   21597   BIC:                         5.897e+05
Df Model:                          15
Covariance Type:            nonrobust
===============================================================================
                  coef    std err          t      P>|t|      [0.025      0.975]
-------------------------------------------------------------------------------
const       -3.692e+07   1.57e+06    -23.516      0.000      -4e+07   -3.38e+07
bedrooms    -3.416e+04   1902.164    -17.961      0.000   -3.79e+04   -3.04e+04
bathrooms    4.234e+04   3164.159     13.381      0.000    3.61e+04    4.85e+04
sqft_living    179.6412      3.643     49.307      0.000     172.500     186.782
sqft_lot         0.1272      0.048      2.636      0.008       0.033       0.222
waterfront    5.879e+05   1.75e+04     33.626      0.000    5.54e+05    6.22e+05
view          4.944e+04   2145.243     23.048      0.000    4.52e+04    5.36e+04
condition       3.1e+04   2347.988     13.202      0.000    2.64e+04    3.56e+04
grade         9.726e+04   2158.970     45.048      0.000     9.3e+04    1.01e+05
sqft_basement   -33.2686      3.937     -8.450      0.000    -40.986     -25.551
yr_built     -2453.0853     70.985    -34.558      0.000   -2592.221   -2313.950
yr_renovated    21.5718      3.676      5.869      0.000      14.367      28.776
lat           5.613e+05   1.05e+04     53.524      0.000    5.41e+05    5.82e+05
long         -1.174e+05   1.19e+04     -9.879      0.000   -1.41e+05   -9.41e+04
sqft_living15   27.3428      3.431      7.969      0.000      20.617      34.068
sqft_lot15      -0.3938      0.074     -5.340      0.000      -0.538      -0.249
==============================================================================
Omnibus:                    18531.230   Durbin-Watson:                   1.993
Prob(Omnibus):                  0.000   Jarque-Bera (JB):          1879952.363
Skew:                           3.616   Prob(JB):                         0.00
Kurtosis:                      48.114   Cond. No.                     5.76e+07
==============================================================================
```